Diabetes Mellitus — Comprehensive Guide

1. Overview

Diabetes mellitus is a group of metabolic diseases characterized by hyperglycemia resulting from defects in insulin secretion, insulin action, or both. Chronic hyperglycemia leads to long-term damage, dysfunction, and failure of various organs, especially the eyes, kidneys, nerves, heart, and blood vessels.

According to the IDF Diabetes Atlas 2021, approximately 537 million adults (20-79 years) are living with diabetes. India alone accounts for over 77 million people with diabetes, making it the second-largest diabetic population in the world, behind China.

The global epidemic is largely driven by urbanization, sedentary lifestyles, unhealthy diets, and obesity. India faces the dual burden of both Type 2 diabetes and malnutrition-related diabetes, particularly in lower-income populations.

2. Classification of Diabetes

Type 1 Diabetes Mellitus

Type 1 DM (formerly insulin-dependent or juvenile-onset diabetes) results from autoimmune destruction of pancreatic beta cells, leading to absolute insulin deficiency. Patients require lifelong insulin therapy to survive.

It accounts for 5-10% of all diabetes cases. Peak onset is in childhood and adolescence, though it can occur at any age. Genetic susceptibility (HLA-DR3, HLA-DR4) and environmental triggers (viral infections) are implicated.

Latent Autoimmune Diabetes in Adults (LADA) is a slowly progressive form of Type 1 that presents in adulthood, often initially resembling Type 2 diabetes. GAD antibody testing helps distinguish it.

Type 2 Diabetes Mellitus

Type 2 DM accounts for 90-95% of all diabetes and results from progressive insulin secretory defect against a background of insulin resistance. The pancreas initially compensates by producing more insulin, but over time, beta cell function declines.

Risk factors include: age >45, obesity (especially central/visceral obesity), physical inactivity, family history, history of gestational diabetes, prediabetes, hypertension, dyslipidemia (low HDL, high triglycerides), polycystic ovary syndrome, and South Asian ethnicity.

South Asians develop Type 2 DM at lower BMIs and younger ages compared to Caucasian populations — a phenomenon related to greater visceral fat deposition and higher insulin resistance at lower body weights.

Gestational Diabetes and Prediabetes

Gestational Diabetes Mellitus (GDM) is diabetes diagnosed in the second or third trimester of pregnancy that was not clearly overt diabetes prior to pregnancy. It affects 10-14% of pregnancies in India. GDM increases the risk of complications for both mother and baby and significantly raises the risk of future Type 2 DM.

Prediabetes includes impaired fasting glucose (IFG: fasting glucose 100-125 mg/dL) and impaired glucose tolerance (IGT: 2-hour OGTT glucose 140-199 mg/dL). It affects 77 million Indians and progresses to Type 2 DM at a rate of 5-10% per year without intervention.

3. Pathophysiology

Insulin, produced by pancreatic beta cells, facilitates glucose uptake by muscle, fat, and liver cells. In Type 2 diabetes, peripheral tissues (especially skeletal muscle) become resistant to insulin's actions, requiring more insulin to achieve the same glucose uptake.

The liver in insulin-resistant states fails to suppress gluconeogenesis appropriately, leading to excess glucose production even in the fed state. Over time, beta cells are unable to compensate for the increasing demand, and hyperglycemia ensues.

Chronic hyperglycemia damages blood vessels and nerves through multiple mechanisms: advanced glycation end-products (AGEs), oxidative stress, activation of protein kinase C, and polyol pathway activation. These mechanisms underlie the classic micro- and macrovascular complications of diabetes.

The 'ominous octet' of Type 2 diabetes pathophysiology (DeFronzo) includes: decreased insulin secretion, increased glucagon secretion, increased hepatic glucose production, decreased incretin effect, increased lipolysis, increased glucose reabsorption by kidneys, decreased glucose uptake in muscle, and neurotransmitter dysfunction.

4. Symptoms and Clinical Presentation

Classic symptoms of hyperglycemia: polyuria (frequent urination), polydipsia (excessive thirst), polyphagia (excessive hunger), unexplained weight loss, fatigue and weakness, blurred vision, slow-healing wounds, and recurrent infections (skin, gums, UTI, candidiasis).

Type 1 DM typically presents dramatically over days to weeks with the classical triad and often with diabetic ketoacidosis (DKA). Type 2 DM often presents insidiously — many are diagnosed incidentally on routine testing or when complications are already present.

Diabetic ketoacidosis (DKA): primarily in Type 1 DM, triggered by insulin deficiency ± a precipitant (infection, missed insulin). Signs: polyuria, polydipsia, nausea, vomiting, abdominal pain, fruity (acetone) breath, Kussmaul breathing (deep rapid), confusion, coma. Blood glucose >250 mg/dL, ketonemia, metabolic acidosis.

Hyperosmolar Hyperglycemic State (HHS): primarily Type 2. Very high glucose (>600 mg/dL), profound dehydration, altered consciousness, without significant acidosis or ketosis. High mortality — requires aggressive fluid replacement.

Hypoglycemia: blood glucose <70 mg/dL. Symptoms: sweating, trembling, palpitations, hunger (adrenergic); confusion, behavior change, seizures (neuroglycopenic). Always carry fast-acting glucose. Severe: loss of consciousness — requires glucagon injection or IV dextrose.

5. Diagnosis

Diagnostic Criteria	Values
Fasting Plasma Glucose (FPG)	≥126 mg/dL (after 8-hour fast)
2-hour OGTT (75g glucose)	≥200 mg/dL
HbA1c	≥6.5% (48 mmol/mol)
Random Plasma Glucose + symptoms	≥200 mg/dL
Prediabetes — IFG	100-125 mg/dL fasting
Prediabetes — IGT	140-199 mg/dL (2-hr OGTT)
Prediabetes HbA1c	5.7-6.4%

In the absence of unequivocal hyperglycemia, results should be confirmed by repeat testing on a different day. HbA1c may be unreliable in hemolytic anemia, hemoglobinopathies (common in India), and in the first and second trimesters of pregnancy.

6. Pharmacological Management

Insulin Therapy

Insulin is essential for Type 1 DM and required by many Type 2 patients over time. Types include: Rapid-acting (aspart, lispro, glulisine — onset 10-15 min), Short-acting (regular insulin — onset 30-60 min), Intermediate-acting (NPH — onset 1-2 hours), Long-acting (glargine, detemir, degludec — near-peakless 24-hour coverage).

Basal-bolus regimen: long-acting insulin once or twice daily (basal coverage) + rapid-acting insulin before each meal (bolus). This most closely mimics physiological insulin secretion and provides the best glucose control.

Insulin pens, pumps (CSII), and newer closed-loop systems (artificial pancreas) improve accuracy and convenience. Continuous Glucose Monitoring (CGM) combined with pumps forms the basis of hybrid closed-loop systems now available in India.

Oral and Non-Insulin Agents

Metformin (500 mg-2550 mg/day): first-line for Type 2 DM. Reduces hepatic glucose production, improves insulin sensitivity. GI side effects (nausea, diarrhea) can be minimized by starting low and taking with food. Avoid in eGFR <30. Unique: does not cause hypoglycemia alone.

Sulfonylureas (glipizide, glibenclamide, gliclazide, glimepiride): stimulate insulin secretion. Risk of hypoglycemia and weight gain. Low cost makes them widely used in India. Gliclazide MR and glimepiride have lower hypoglycemia rates.

SGLT2 inhibitors (empagliflozin, dapagliflozin, canagliflozin): cause glycosuria, reduce BP, weight. Cardiovascular and renal benefits beyond glucose control (EMPA-REG, CREDENCE trials). Risk of genital mycotic infections, DKA in Type 1.

GLP-1 Receptor Agonists (semaglutide, liraglutide, dulaglutide): injectable (weekly or daily). Reduce glucose, weight, and cardiovascular events. Semaglutide now available orally. GI side effects common initially.

DPP-4 inhibitors (sitagliptin, vildagliptin, teneligliptin): well-tolerated, weight-neutral, low hypoglycemia risk. Modestly effective. Saxagliptin associated with heart failure hospitalization.

Pioglitazone: improves insulin sensitivity. Risk of fluid retention, heart failure, and fractures. May reduce non-alcoholic fatty liver disease. Not recommended with bladder cancer.

7. Monitoring and Targets

Parameter	Target
HbA1c (most patients)	<7.0% (<53 mmol/mol)
HbA1c (elderly/comorbid)	<7.5-8.0%
Fasting Glucose	80-130 mg/dL
Post-meal Glucose (2-hr)	<180 mg/dL
Blood Pressure	<130/80 mmHg
LDL Cholesterol	<70 mg/dL (high CV risk)
Triglycerides	<150 mg/dL
BMI	18.5-22.9 kg/m² (Asian target)

Self-monitoring of blood glucose (SMBG): Type 1 — 4-7 times daily. Type 2 on insulin — as directed. Type 2 on oral agents — fasting and 2-hour post-meal periodically. CGM is increasingly accessible and offers more complete glucose pattern data.

8. Complications

Microvascular Complications

Diabetic Retinopathy: leading cause of new blindness in working-age adults. Non-proliferative (NPDR) → Proliferative (PDR) → maculopathy. Annual dilated eye exam mandatory. Treatment: laser photocoagulation, anti-VEGF injections (ranibizumab, bevacizumab).

Diabetic Nephropathy: starts with microalbuminuria (30-300 mg/day), progresses to macroalbuminuria and declining GFR. Leading cause of end-stage renal disease in India. Control BP, use ACE inhibitor/ARB, restrict protein intake, control glucose tightly.

Diabetic Neuropathy: distal symmetric polyneuropathy (burning, tingling, numbness in stocking-and-glove distribution), autonomic neuropathy (gastroparesis, erectile dysfunction, orthostatic hypotension, cardiac autonomic neuropathy), mononeuropathies.

Macrovascular Complications and Foot Care

Cardiovascular disease is the leading cause of death in Type 2 DM. Annual 10-year CVD risk assessment. Use SGLT2 inhibitors or GLP-1 agonists with proven CV benefit in high-risk patients. High-intensity statin therapy for all over 40.

Diabetic Foot: neuropathy + ischemia + infection = foot ulcer and amputation risk. Daily foot inspection. Proper footwear. Prompt treatment of even minor wounds. Refer to wound care specialist for non-healing ulcers. Multidisciplinary foot team reduces amputation rates by 50-80%.

9. Emergency Management

DKA management (hospital): Fluid resuscitation — 0.9% NaCl 1L over first hour, then 250-500 ml/hr based on hydration. IV insulin infusion (0.1 units/kg/hr). Potassium replacement (never give insulin without checking potassium). Identify and treat precipitant. Monitor glucose hourly, electrolytes 2-4 hourly.

HHS management: Slower fluid replacement (more volume, over 24-48 hrs). Goal is gradual glucose reduction. Insulin only when fluids alone fail to reduce glucose adequately. High DVT risk — prophylactic anticoagulation.

Hypoglycemia: Conscious patient — 15g fast-acting carbohydrate (5 glucose tablets, 150 ml juice, 3 teaspoons of sugar dissolved in water). Repeat in 15 min if still low. Follow with snack. Unconscious patient — do NOT give anything by mouth. IV dextrose 25-50 ml of 50% dextrose. IM/SC glucagon 1 mg if no IV access.

10. Prevention and Diabetes in India

Type 1 DM currently has no known prevention. Type 2 DM can be largely prevented or delayed. The DPP (Diabetes Prevention Program) trial showed that intensive lifestyle modification (5-7% weight loss, 150 min/week moderate exercise) reduced progression from prediabetes to diabetes by 58% — more effective than metformin.

India-specific challenges: high rates of thin-fat phenotype (metabolic obesity at normal BMI), early onset, rapid urbanization, cost of CGM and newer medications, limited access to dietitians and diabetes educators, and high proportion of undiagnosed diabetes.

Government of India initiatives: National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke (NPCDCS). Free metformin and basic insulin available at government hospitals. Pradhan Mantri Jan Aushadhi Kendras offer affordable generic medications.

The future of diabetes care in India lies in community screening programs, digital health interventions (apps for SMBG logging, teleconsultation), point-of-care HbA1c testing, and affordable CGM to improve glycemic outcomes at the population level.